BGIN Block #9 Meeting Report

- Rebuilding Value, Governance and Reputation on Blockchain -

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The Global Network for Blockchain Stakeholders™

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IKPWG Interactive Workshop Session 1: ZKP and its Applications

Date: Nov 21, 2023
Time: 13:45 - 15:15
Session Chair: Leona Hioki (Intmax)

1. Paper Summary

- The report provides a comprehensive overview of Zero-Knowledge Proofs (ZKP) and their practical implementations. Here's a summarized overview of the content:
    a. Definition and History of ZKP: The report begins by defining Zero-Knowledge Proofs (ZKP), discussing their history, characteristics, and classification.
    b. ZKP and Society: This section explores the relationship between ZKP and societal aspects, including programming languages and libraries exclusive to ZKP, performance trade-offs, historical vulnerabilities, and various applications of ZKP.
    c. Technical Aspects of ZKP: The document delves into the technicalities of ZKP, including their composition, characteristics, and specific types like zkRollup, along with their history and concerns.
    d. Recursive zkSNARKs: This section explains the concept of recursive zkSNARKs, highlighting their significance, applications, and challenges.
    e. Classification of NIZKs: Various types of practical NIZKs, such as zkSNARKs and zkSTARKs, are classified and discussed, including their development history and specific characteristics.
    f. Frontend and Backend Aspects of ZKP: Different aspects of ZKP are explored, including frontend (circuit design, constraint systems) and backend (pairing, polynomial commitment schemes) components.
    g. Libraries and Languages for ZKPs: The report lists and describes various programming languages and libraries used for implementing ZKP.
    h. Performance, Vulnerabilities, and Applications of ZKP: The document assesses the performance criteria for ZKPs, discusses potential vulnerabilities, and provides examples of real-world applications, such as cryptocurrencies and privacy-focused technologies.
    i. zkRollup: This section offers a detailed look at zkRollup, its classification, history, and concerns related to its implementation and use.
    j. Regulatory, Security, and Privacy Considerations: The report concludes with discussions on regulatory, security, and privacy considerations.

Link to the paper summary: BGIN-ZKPdoc

2. Discussion points

- Definition and Description of ZKP
    - Suppose there is a data lake that stores licenses, and some exchanges can share and crawl the license data, then the exchange proves that "I posess the license" without disclosing its license.
    - The difference between ZKP and FHE or other cryptography is that ZKP is a special Cryptography, but not Encryption.
- The traditional misconception about ZKP
    - With keywords like zkEVM, zkRollup, ..., etc. flying around, we feel very much that the definition, purpose and requirements are vague.
    - zkRollup focuses on scalability, but zk-application mainly focuses on privacy. Requirements for using ZKP vary depending on the purpose, but the public lumps them together.
    - Because of the gap between what can be constructed and what is available when considering legal processes, there is concern that on the one hand, technology will make it safe to use while protecting privacy, and on the other hand, analog methods using email, paper, and PDFs will accelerate. As a result, it would hinder the promotion of innovation.
    - ZKP performance is still in its infancy, so some user experience sacrifices must be made to protect privacy
- Sharing some interesting use cases about ZKP
- Developer Experience about ZKP
    - Currently, writing zkCircuit requires additional mathematical knowledge apart from existing web coding.
    - It isn't fast enough.
    - In some cases, users need to download certain files to their own desktops in order to generate a Proof.
    - Few documentations
    - To translate a statement to circuit, you need to dig through the GitHub documentation

3. Next Steps

- Propose and discuss real-world challenges and potential applications of ZKP to those challenges, not limited to zkSNARK, but also VC, etc.

- Promote feedback and discussion about the ZKP study report in the Discourse community.